Ivan Jimenez Pineda

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EDUCATION

Harvey Mudd College - Claremont, CA
Bachelor of Science in Engineering | GPA: 3.384 | | Expected Graduation: May 2028 |

PROFESSIONAL EXPERIENCE

CRPA Characterization Clinic Project - The Aerospace Corporation / Federal Aviation Administration | September 2026 - Present |

- Designed and machined custom **electromechanical mounting fixtures** for positioning turntables, designing the hardware so the **Controlled Reception Pattern Antenna** may be placed securely inside the **anechoic chamber** for testing.
- Lead the automation of the AntRunner antenna positioner that orients the CRPA to commanded angles inside the anechoic chamber, owning its electrical, software, and mechanical setup.
- Develop the **Python control script** that drives the positioner's rotors and deploy it to the AntRunner's **ESP32 controller** from a **Raspberry Pi**.

Mechanical Engineering Intern - ALLFAST Fastening Systems (Penn Aero) | May 2026 - August 2026 |

- Created **65 tooling layout drawings** and modeled parts for tooling assemblies using **SolidWorks** to support over **224 million parts** from the production backlog.
- Updated **BOMs** in the **ERP** system with accurate tooling numbers and quantities, enabling proper tracking and forecasting to mitigate heading downtime as a result of tooling shortages to allow for tools to be ordered quickly through **MRP**.
- Validated tooling layouts and BOM updates by physically assembling tooling and conducting reviews with engineering teams and machine operators, ensuring accuracy and developing strong technical writing and documentation skills.

Machine Shop Proctor / Workshop Proctor - Harvey Mudd College | May 2025 - Present |

- Instruct students on the safe operation of machining equipment (**Manual Mill, Lathe, CNC Router**), emphasizing adherence to process specifications and **tolerances down to 0.001"**.
- Train students to interpret technical drawings and apply **GD&T** to **optimize production processes** and reduce fabrication errors.
- Guide students through manual machining and metalworking processes to safely fabricate custom projects, including machined pens, rings, spinning tops, and sheet metal flowers.

PROJECTS

Autonomous Underwater Vehicle | August 2025 - May 2026 |

- Engineered and fabricated a **low-cost** autonomous **electro-mechanical robot** featuring a custom **winch mechanism** to deploy a tethered sensor payload for depth-resolved environmental data collection.
- Designed analog signal conditioning circuitry, utilizing **unity gain and inverting offset operational amplifiers**, to interface an **NTC thermistor**, **pressure sensor**, and **amorphous silicon solar cell** with a **Teensy microcontroller**.
- Programmed the embedded system in C for synchronized sensor logging and utilized **MATLAB** to calibrate raw data, capturing temperature and luminosity fluctuations relative to depth during coastal field deployments at Dana Point Harbor.

Brachistochrone Curve Model | February 2025 - May 2025 |

- Modeled cycloid, linear, and parabolic descent paths in **SolidWorks** using the **Equation Driven Curve** feature to optimize geometric accuracy and demonstrate kinematic principles for a client-requested physics demonstration.
- Fabricated the primary ramp structures and custom corner fixtures using a **CNC Router,** ensuring precise assembly and tolerances.

MITES Summer Scholar Symposium Project | June 2023 - July 2023 |

- Co-developed "Project Beatzthoven," an interactive **machine learning application** built in **Python** that algorithmically generates classical music utilizing Markov Chain logic.
- Applied **systematic debugging** practices and software engineering fundamentals to process data structures and balance competing project priorities within a strict 6-week development cycle to present at an MIT showcase.

The Machinist's Hammer | March 2025 - April 2025 |

- Machined a multi-material alloy steel hammer including a hard face, soft face, handle, and a stand (**AISI 4340 and AISI 1015**).
- Applied heat treatment (**857°C)** to hard face and verified results using a Rockwell hardness tester (**RC 43-48**).
- Operated a mill, lathe, vertical bandsaw, belt sander, sand blaster, laser cutter, and 3D printer.

SKILLS

Programming: Python, C, SystemVerilog, LabVIEW, MATLAB, HTML, Arduino IDE, Data Acquisition (MyDAQ), VNA Analysis Oscilloscope Testing, Git, Github, FPGA Design, CSS, Raspberry Pi, ESP32
Manufacturing: Precision Machining (Mill and Lathe), CNC Router, Laser Cutter, Waterjet, Heat Treatment, Rockwell Hardness Testing, Sand Blasting, GD&T, SolidWorks, 3D Printing, FEA, Fusion 360 CAM, Soldering
Languages: Fluent in English and Spanish